CS 4500-Software Engineering

SG2: Wandering in the Woods

Test Plan Document

Group Number: 2

Team Members:

- **Alex Fahnestock**
- **AJ Soma Ravichandran**
- **May Salahaldin**
- **Tressa Millering**
- **Jackie Herbstreit**

Version: 1.0
Date of First Design: April 2, 2026

Table of Contents

Title Page ……..……... 1

Table of Contents ……...…….. 2

Test 1 ……..……..…… 3

Test 2 ……..……..…… 4

Test 3 ……..……..…… 5

Test 4 ……..……..…… 6

Test 5 ……..……..…… 8

Test 6 …… 9

Test Case 1

Test Description: Invalid input for N

User enters: -1, 1000, 0.5, twenty, 50

Expected Behavior:

After the program explanation, the program prompts the user to enter **N**.

- If the value is too small, the program displays:
  "Error: Value is too small. Minimum is 2."
  Then, it reprompts the user for **N**.
- If the value is too large, the program displays:
  "Error: Value is too large. The maximum is 100."
  Then, it reprompts the user for **N**.
- If a floating-point number is entered, the program displays:
  "Error: Floating-point input. Please enter a whole number."
  Then, it reprompts the user for **N**.
- If a non-numeric value is entered, the program displays:
  "Error: Non-numeric input. Please enter a whole number."
  Then, it reprompts the user for **N**.

After a valid value for **N** is entered, the program prompts the user to enter **T**.

Actual Result:

```
Shell ×   Program tree ×
and a detailed report will be saved to a text file.

-------------------------------------------------------
            Enter values for the simulation:
-------------------------------------------------------
Enter grid size N (2-100): -1
Error: value is too small. Minimum is 2.
Enter grid size N (2-100): 1000
Error: value is too large. Maximum is 100.
Enter grid size N (2-100): 0.5
Error: Floating point input. Please enter a whole number.
Enter grid size N (2-100): twenty
Error: Non-numeric input. Please enter a whole number.
Enter grid size N (2-100): 50
Enter time limit T (2-1000000):
```

Pass/Fail: Pass
Date: 04/09/2026
Tester: Tressa Millering

Test Case 2

Test Description: Invalid input for T

User enters: 2, -1, 1000001, 0.5, twenty, 50

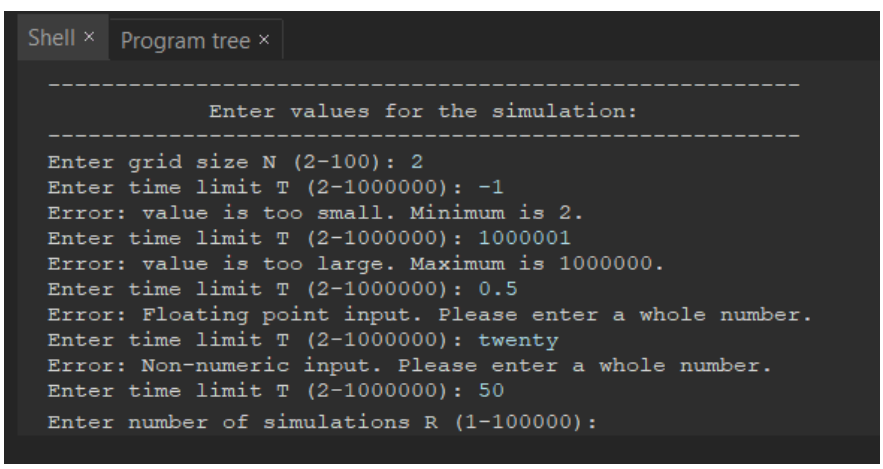
Expected Behavior:

After the program explanation, the program prompts the user to enter **N**, followed by **T**.

- If the value of **T** is too small, the program displays:
  "Error: Value is too small. Minimum is 2."
  Then, it reprompts the user for **T**.
- If the value of **T** is too large, the program displays:
  "Error: Value is too large. The maximum is 1,000,000."
  Then, it reprompts the user for **T**.
- If a floating-point number is entered, the program displays:
  "Error: Floating-point input. Please enter a whole number."
  Then, it reprompts the user for **T**.
- If a non-numeric value is entered, the program displays:
  "Error: Non-numeric input. Please enter a whole number."
  Then, it reprompts the user for **T**.

After a valid value for **T** is entered, the program prompts the user to enter **R**.

Actual Result:



Pass/Fail: Pass
Date: 04/09/2026
Tester: Tressa Millering

Test Case 3

Test Description: Invalid input for R

User enters: 2, 2, -1, 100001, 0.5, twenty, 50

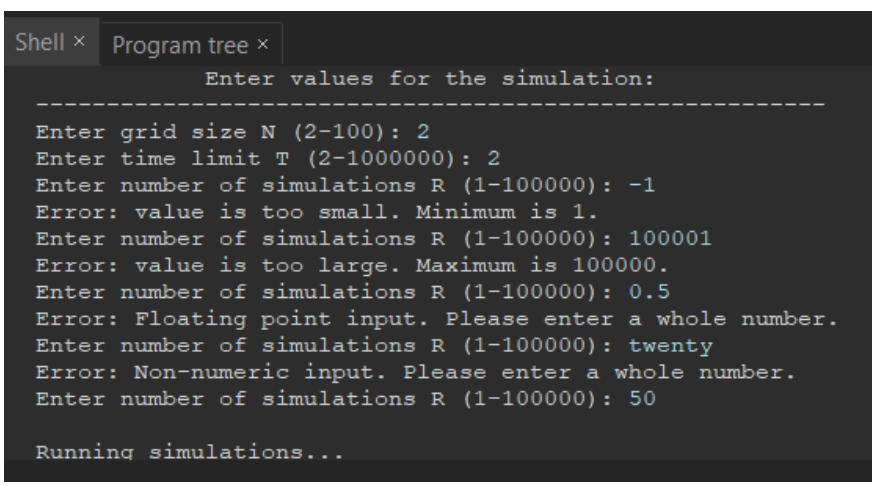
Expected Behavior:

After the program explanation, the program prompts the user to enter **N**, **T**, and then **R**.

- If the value of **R** is too small, the program displays:
  "Error: Value is too small. Minimum is 2."
  Then, it reprompts the user for **R**.
- If the value of **R** is too large, the program displays:
  "Error: Value is too large. The maximum is 100,000."
  Then, it reprompts the user for **R**.
- If a floating-point number is entered, the program displays:
  "Error: Floating-point input. Please enter a whole number."
  Then, it reprompts the user for **R**.
- If a non-numeric value is entered, the program displays:
  "Error: Non-numeric input. Please enter a whole number."
  Then, it reprompts the user for **R**.

After a valid value for **R** is entered, the program proceeds to run the simulations.

Actual Result:



Pass/Fail: Pass

Date: 04/10/2026

Tester: Tressa Millering

Test Case 4

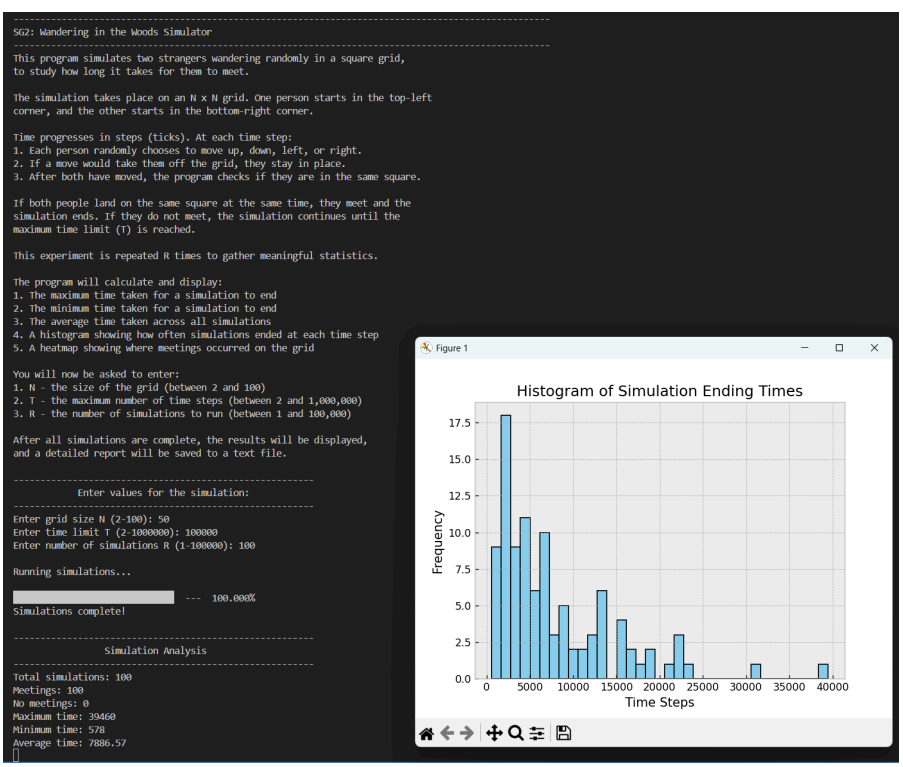
Test Description: Test the program with higher input values for **N**, **T**, and **R**. Verify that the simulation function correctly processes the data, generates the histogram and heatmap visualizations, and writes the results to the output text file.

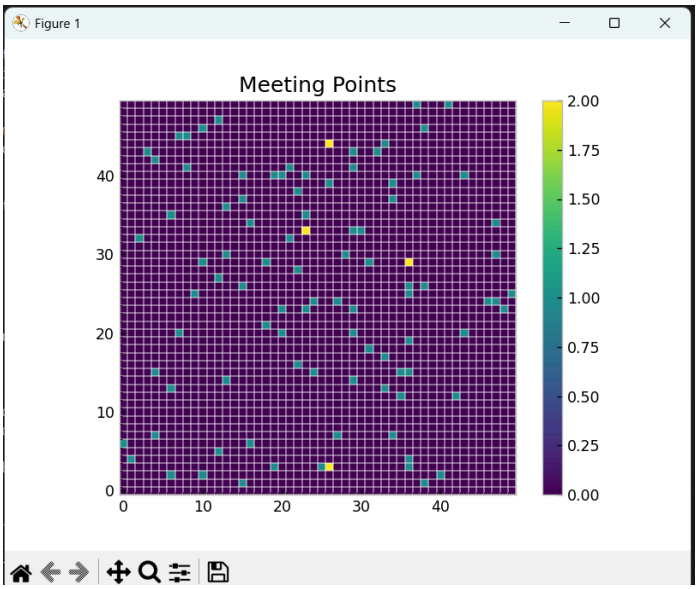
User enters: N = 50, T = 100000, R = 100

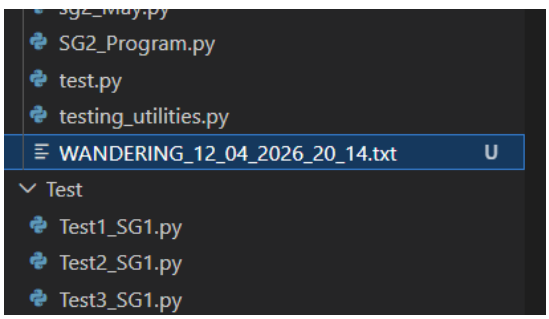
Expected Behavior:

- After the program explanation, the program prompts the user to enter values for **N**, **T**, and **R**. Once the values are entered, the program validates them and, if they are valid, proceeds to run the main simulation.
- While the simulation is running, the program displays a loading progress bar from **0% to 100%** to indicate progress.
- After the simulation is complete, the program displays the results including Total simulations, Number of meetings, Number of no meetings, Maximum time, Minimum time, Average time
- Additionally, the program displays a histogram visualization if a meeting occurred and a heatmap visualization
- Finally, the program writes the results to a text file using the correct timestamp format: **WANDERING_DD_MM_YYYY_HH_MM.txt**

Actual Result:







Pass/Fail: Pass

Date: 04/12/2026

Tester: AJ Soma Ravichandran

Test Case 5

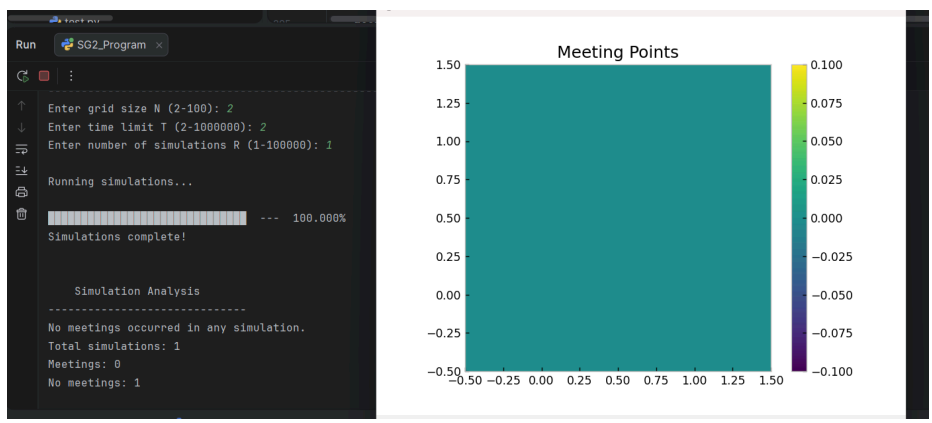
Test Description: Test the program with lowest possible input values for **N**, **T**, and **R**. Verify that the simulation function correctly processes the data, generates the histogram and heatmap visualizations, and writes the results to the output text file.

User enters: N = 2, T = 2, R = 1

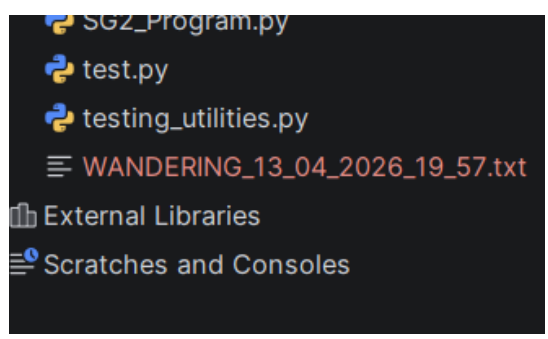
Expected Behavior:

- After the program explanation, the program prompts the user to enter values for **N**, **T**, and **R**. Once the values are entered, the program validates them and, if they are valid, proceeds to run the main simulation.
- While the simulation is running, the program displays a loading progress bar from **0% to 100%** to indicate progress.
- After the simulation is complete, the program displays the results including Total simulations, Number of meetings, Number of no meetings, Maximum time, Minimum time, Average time
- Additionally, the program displays a histogram visualization if a meeting occurred and a heatmap visualization
- Finally, the program writes the results to a text file using the correct timestamp format: **WANDERING_DD_MM_YYYY_HH_MM.txt**

Actual Result:



Pass/Fail: Pass
Date: 04/13/2026
Tester: Tressa Millering



Test Case 6

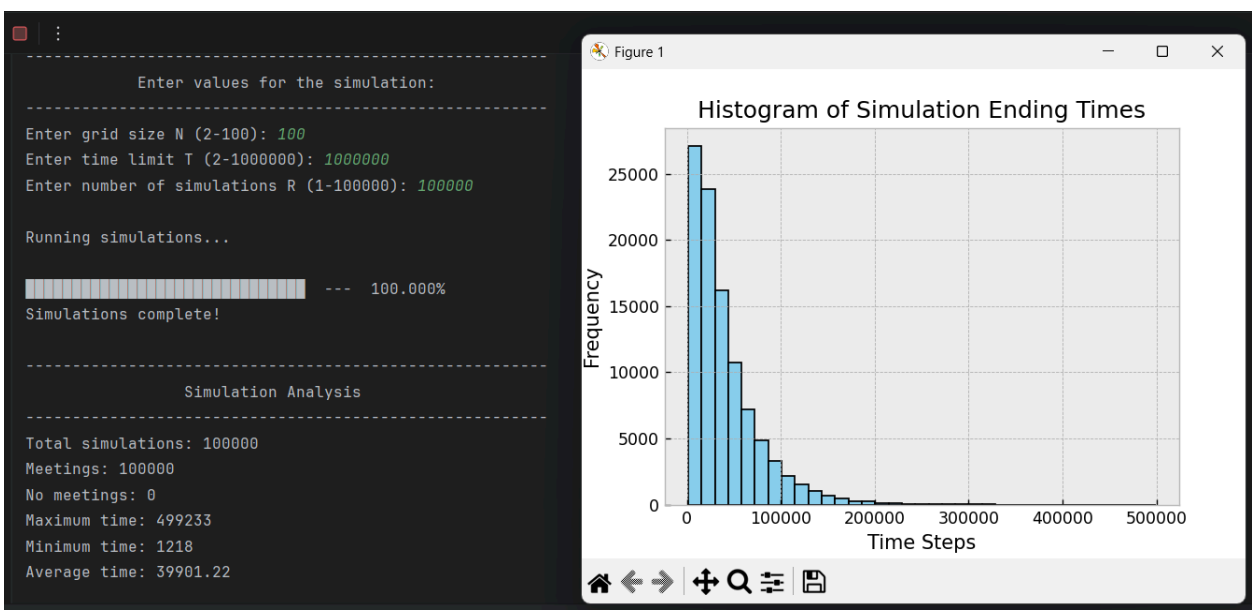
Test Description: Test the program with highest possible input values for **N**, **T**, and **R**. Verify that the simulation function correctly processes the data, generates the histogram and heatmap visualizations, and writes the results to the output text file.

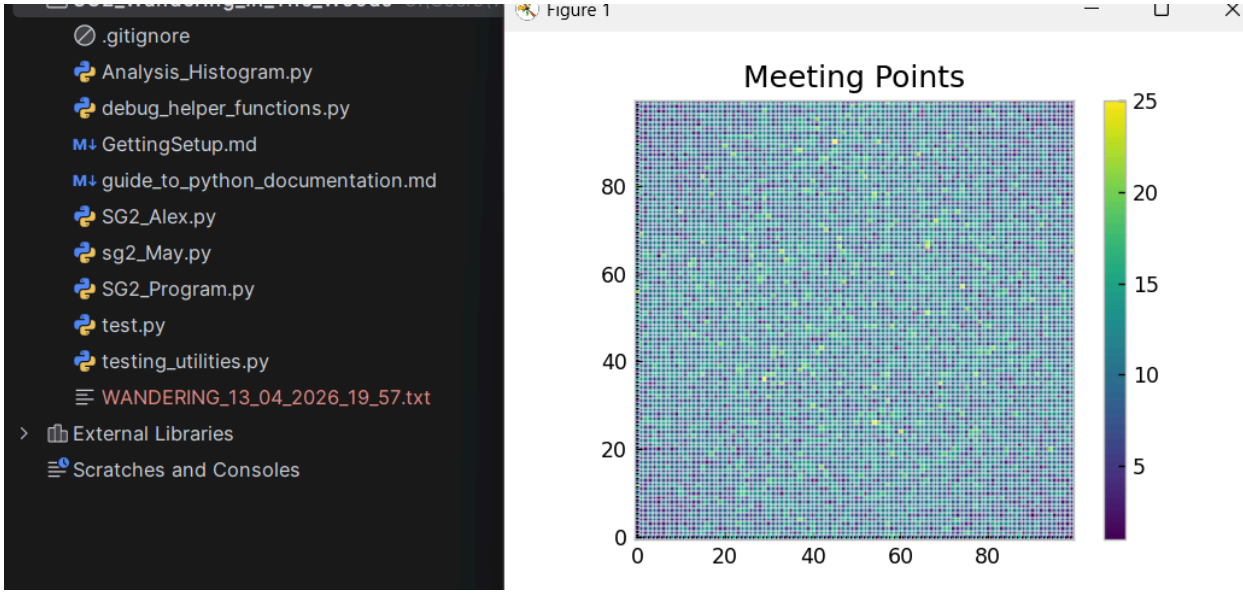
User enters: N = 100, T = 1000000, R = 100000

Expected Behavior:

- After the program explanation, the program prompts the user to enter values for **N**, **T**, and **R**. Once the values are entered, the program validates them and, if they are valid, proceeds to run the main simulation.
- While the simulation is running, the program displays a loading progress bar from **0% to 100%** to indicate progress.
- After the simulation is complete, the program displays the results including Total simulations, Number of meetings, Number of no meetings, Maximum time, Minimum time, Average time
- Additionally, the program displays a histogram visualization if a meeting occurred and a heatmap visualization
- Finally, the program writes the results to a text file using the correct timestamp format: **WANDERING_DD_MM_YYYY_HH_MM.txt**

Actual Result:





Pass/Fail: Pass
Date: 04/13/2026
Tester: Tressa Millering